CN Practical 13: Create a Hybrid Network Using Bluetooth

Aim: To create and demonstrate a hybrid network using Bluetooth along with wired or Wi-Fi networking technologies.

Theory:

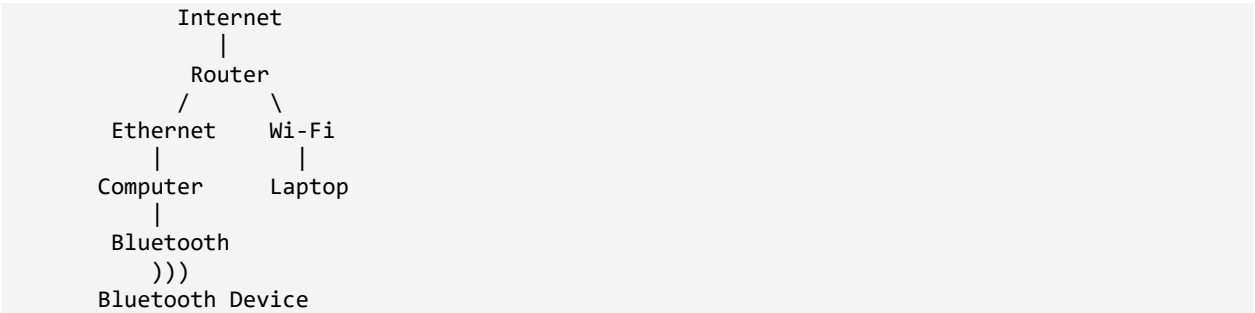
A hybrid network is a network that combines two or more types of network technologies or connection methods. It may include wired Ethernet, Wi-Fi and Bluetooth connections. Different technologies are selected according to their range, speed, power consumption and intended application.

Bluetooth is a short-range wireless communication technology commonly used for connecting phones, laptops, keyboards, headphones and other peripherals. Wi-Fi provides wireless LAN connectivity, while Ethernet provides wired network communication. Combining these technologies allows devices to communicate using the method most suitable for them. For example, a computer may connect to a router through Ethernet, a laptop may use Wi-Fi, and a Bluetooth device may exchange files with a nearby computer.

Materials required:

- Computer or laptop
- Bluetooth-enabled phone or device
- Wi-Fi router or access point
- Ethernet cable
- Bluetooth adapter, if required

Network diagram:



Procedure:

1. Switch on the computer and networking devices.
2. Connect the computer to the router using an Ethernet cable.
3. Connect a laptop or phone to the same network using Wi-Fi.
4. Enable Bluetooth on the computer and the Bluetooth-enabled device.
5. Make the devices discoverable and search for nearby devices.
6. Select the required device and complete the pairing process.
7. Transfer a test file between the paired devices, if supported.
8. Verify internet connectivity on the computer and Wi-Fi device.
9. Observe the differences between Ethernet, Wi-Fi and Bluetooth communication.
10. Record the results of the connectivity and file-transfer tests.

Observation:

The computer communicated with the router through Ethernet, while the laptop or phone connected using Wi-Fi. Bluetooth was used for short-range communication between paired devices. The practical demonstrated how different connection technologies can be combined within a hybrid networking environment.

Conclusion:

A hybrid network using Bluetooth, Wi-Fi and wired Ethernet was demonstrated successfully. The practical explained the roles and applications of different communication technologies. It also highlighted how hybrid networks provide flexibility for connecting various devices.